



MTP Department
SIMHADRI

Note No :1



Ref:SMTPP/MAINT. PLNG./2025-26/MTP/547276

Date:17/01/2026(19:57:30)

Subject:LMI ON SYSTEM FOR PLANT MODIFICATIONS

NTPC Limited

SIMHADRI SUPER THERMAL POWER STATION

MAINTENANCE PLANNING DEPARTMENT

Sub:- Approval of LMI on "System for Plant Modifications"

LMI on "System for Plant Modifications" is reviewed based on **LMI/OD/GEN/006 REV-5, March 2023 & COS-ISO-00-OD/GEN/006 Rev. No.:3, Dec-2020** and **no changes are proposed.**

Format revised as per LMI/OIN/GEN/002 Rev: 04 on Guidelines for Issue of LMIs as a Controlled Document

Submitted for approval.

Encl:

1. LMI for approval at Annexure-I
2. LMI/OD/GEN/006 REV-5, March 2023 at Annexure-II
3. COS-ISO-00-OD/GEN/006 Rev. No.:3, Dec-2020 at Annexure-III

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May pl review

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May Kindly be approved.

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Note No:5

LMI is in line with the OD.

Forwarded for further review.

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LMI is in line with the OD. May please be approved.

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Note No:11

Flow diagram at page 9 of the proposed LMI document (adopted from COS-ISO-00-OD/GEN/006 Rev. No.:3. dated Dec 2020) has been corrected by replacing GM(O&M) in place of AGM(O&M) .

Forwarded for kind approval pl

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NTPC Ltd

SIMHADRI SUPER THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION

LMI/OD/GEN/006

ISSUE NO.2 REVISION NO. 5, Mar 2023

System for Plant Modifications

Approved for Implementation by: **GGM (SIMHADRI)**

Date_____

OPERATION & MAINTENANCE DEPARTMENT
(Energy & Efficiency Management Group)

LMI Name	System for Plant Modifications	
LMI No.	LMI/OD/GEN/006	
Issue No. 2	Revision: 5	Mar 2023

	Name (Sh)	Designation
Prepared by	K.V.K. Bhaskar	DGM (MTP)
Reviewed by	RAJESH CHAWLA	AGM(MTP)
Reviewed by	D.P. PATRA	GM (O&M)

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SYSTEM FOR PLANT MODIFICATIONS

1. INTRODUCTION

This LMI is issued to comply with the operation directive COS-ISO-00-OD/GEN/006, Rev. 3. The LMI specifies the detail procedure of registration and technical assessment of all proposed modifications of Simhadri Super Thermal Power Station as per their intended purpose for safety of the personnel and power plant equipment. This LMI is prepared to lay down the specific procedures for modification as per the operation directives issued by COS.

2. SCOPE

- 2.1. This LMI shall be applicable to all modifications (or design change) to 4 x 500 MW Equipment / Systems / Components of Simhadri Station after the date of issue of this LMI. For modifications made prior to issue of this revision, procedure described at clause no 5.7 is to be followed.
- 2.2. The LMI shall not be applicable to modifications/ design change to be carried out under contractual obligations of the Contractor under the contract irrespective of the period when such modification is carried out. However, in such cases, where modification is necessarily to be carried out by Simhadri before expiry of the warranty period or before PG Test, whichever is later, written consent of the contractor to such proposed modification should be obtained before carrying out the Modification. In all such cases, an approval from Engineering shall be taken, which shall be routed through Regional OS (ROS) and Corporate OS (COS).
- 2.3. The reference design or datum for each Plant / Equipment/ System / Component of the Power Station shall be as per the last document approved by Engineering or the original design drawings.
- 2.4. The LMI shall not be applicable to 'defects' or 'repairs' which require equipment to be reinstated to the original condition.
- 2.5. For the purpose of this LMI, System or Component shall also include Load Bearing Structures

3. SUPERSEDED DOCUMENT

LMI/OD/GEN/006 Rev No.4 Date : Feb 2021

4. RECORDS OF REVISION

Proforma attached at Annexure-I has been partially modified in view of E files.
One point added in Clause 5.2, need for modification.

5. PROCESS

5.1. Definition

5.1.1. For the purpose of this LMI, the term modification or design change shall have the following meaning or interpretation: -

5.1.1.1. 'Modification' shall include

- i) A repair or replacement of a system/ equipment/ component where it involves a change to its reference design or datum, geometry or materials.
- ii) Any change to the settings, or mode of operations, of a component in a control, interlock or protection system (including computer software).
- iii) Any addition or deletion of a system, component, or of a complete system or complete equipment.

5.1.1.2. The following shall not be construed as modifications for the purpose of this LMI.

- i) The repair/ replacement/ modification of **minor components** or **equipment** as may be specified by the **Head of the Station SIMHADRI** in this revised Location Management Instruction. However, details of these shall be intimated to COS.
- ii) The overriding of any interlock (Protection by pass) or any change to the settings, or mode of operation, of any component in a control, interlock or protection system (including computer software), which is authorized, implemented and recorded in accordance with a separate LMI, provided that no such override or change remains in place for a period exceeding 72 hrs. If it is to exceed 72 hrs, it will be treated a modification and will require registration and assessment in accordance with this LMI.

5.2. Need for Modification

The need for modification or design change may arise from:

- Recommendations of an enquiry report/ ROPR/ ORT/ Audits.
- Proven best practice elsewhere in the utilities known globally.
- Improvement in efficiency / commercial parameters.

- Improvement from perspective of Energy efficiency, Safety, Health and Environment
- Statutory / Legal requirements
- Feedback from major failures / trip analysis.
- Constraints faced during Plant Operation.

5.3. Proposal

5.3.1. In order to keep record of all modifications or design change to Plant / Equipment's / Systems / Components, all proposals for modifications shall be registered with the Station **MTP department**. Further, to ascertain the suitability of the modifications, all proposals will be assessed technically as per the system laid down hereunder:

5.3.2. A format for Registration of all proposals for modification or design change is given at Annexure I (Modification Registration Performa / Design change Request).

5.3.3. The proposal for modification / design change may be initiated by concerned department at station and sent to Head of (O&M) of the station for formal initiation. The proposal shall give sufficient details of modification i.e name of the Plant / Equipment / Component / System, its Location, Name of the Supplier / Manufacturer, reasons for modification, approving authority etc. The details of modification shall be maintained by the MTP department of the Power Station in the Performa (request for design change) given at Annexure – 1. A flow diagram showing the process of the system for plant modification / design change has been given in Annexure –II.

5.4. Categorization of Modifications

The category of technical assessment required i.e. internal or external to the station, shall be approved by **ED (Southern Region)** based on the note giving details of modifications received from the **Head of Power Station Simhadri**.

5.4.1. For the purpose of LMI technical assessment of proposals shall be of two types.

- a). Category A (requiring external assessment).
- b). Category B (requiring internal assessment).

5.4.2 Category A assessment shall be required in case of all proposals which, if inadequately conceived or executed, could lead to the following:

- A major safety / environmental implication.
- A major plant failure (Boiler, Turbine, Generator, Transformer, Cooling Water System, Ash Disposal System, Coal handling Plant, Switch yard)

5.4.2. Category A assessment shall also be required in case of

- Any change in the protection scheme for main equipment i.e. Boiler, Turbine, and Generator.
- Modifications related to improvement of efficiency / commercial parameters of major equipments.

5.4.3. Category 'A' assessment may not be required:

- i) If repair/replacement of any System or Component or Equipment is carried out without changing the original design or datum and using NTPC approved procedures relevant to such repair.
- ii) If replacement of any system or component or equipment is being done due to age or obsolescence by a modern equivalent with no change in original design or datum, materials or operating environment.

5.5. Category 'A' Assessment

The proposals for modification of all Category 'A' proposals as defined above shall be submitted for technical approval to Engineering by the Power Station through ROS & COS. Engineering may seek the comments from other departments of Corp. Centre e.g. R&D, Safety, CENPEEP etc. The technical approval from engineering will be sent to the Station with a copy to COS and ROS. Thereafter, the proposal would be submitted for approval of Competent Authority by Power Station as per DOP. If required, detailed engineering will be done by engineering and sent to the station for implementation of the modification under intimation to COS and ROS.

5.6. Category 'B' Assessment

The proposals which do not qualify for category A assessment shall be categorized as Category B. The proposals involving category B assessment shall be scrutinized and assessed by a team comprising representative of O&M, FQA and Technical Services Departments at SIMHADRI Station in consultation with Corp. OS & Engg. and approved in accordance with DOP if any costs are involved. After the

modification has been done a post facto approval will be sought by the station in charge from Engineering detailing the requirement for the modification, implementation status etc. Any recommendations from engineering shall be implemented by the Station.

5.7. Modification made prior to this LMI

The station shall compile all the modification with relevant details made prior to the issue of this document. For all the modifications which have not been vetted or approved by Engineering, the station will be required to intimate the same to Engineering through ROS and COS along with the relevant details, pre and post modification assessment and amendment to documents and drawings wherever applicable. Any recommendations from engineering shall be implemented by the Station.

5.8. Amendment of Station Documentation

All relevant as built drawings, operation and maintenance instructions, details in the stores catalogue and other requirements for modified Equipment/ System / Component will be amended before the first modified system or component is commissioned. The amended documentation / as built drawings must be approved by the same authority that initially approved the documentation / drawing. The master List of documents and drawings must accordingly be revised. The issue of these modified documents as above shall be controlled and the old version are stamped as superseded. Where the modification is to be progressive, covering various areas of the Power Station, then care must be taken to clearly identify the system or component or equipment to which the amended documentation applies. The necessary records will be updated in Modification Register Performa.

5.9. Execution of the Plant Modification

After the receipt of the detailed Engineering from Corp. Centre, MTP will find out if any procurement is required or not. If any procurement is required than concerned maintenance department will initiate the procurement. After receipt of the material it will execute the modification with relevant quality checks by FQA. Commissioning will be done by Operation. Reports, as required, will be generated and given to MTP for consolidation.

5.10. Post Modification Assessment

After the system or component has been commissioned, an assessment shall be carried out by **Head of O&M** at the Power Station to determine-

- i) that the modification has been implemented in accordance with the approval
- ii) the resulting change in performance as compared to that proposed originally
- iii) that the integrity of the plant safety has not been impaired.

5.10.1 Where appropriate, this assessment shall include such tests and trials specified in the approval.

5.10.2 A report shall be generated by the **Head of O&M** of the Station stating the improvement after the modification with respect to pre-modification status and shall be sent to Head of Corporate OS and Regional OS. Final acceptance of the modification shall be given by COS in association with Engineering. In case of non acceptance of the modification necessary analysis shall be done by **MTP** along with the concerned department(s). Either corrective action is taken and post modification assessment carried out; or pre-modification status is restored. Site may also initiate a fresh proposal for modification.

5.10.3 The Modification Register Performa will now be updated completely and shall be circulated to all concerned including Corp. OS.

5.10.4 Compliance of LMI

The compliance of this LMI shall be verified during the course of Internal and External LMI Audits and Technical Audit of the Station by Corporate OS.

5.10.5 Records

The following records shall be maintained by MTP department at the station.

- Records of modifications as per Modification Registration proforma.
- Records of approval of modification.
- Reports of Post Modification Assessment.

6 REVIEW

This document shall be reviewed by Head of Project every two years or as necessary.

ANNEXURE - I

MODIFICATION REGISTER PROFORMA / DESIGN CHANGE REQUEST

(can be maintained in soft copy at designated place as per LMI made on this OD)

MODIFICATION REGISTER	Serial No.:
For Plant, Apparatus and Structure	Date Proposed:

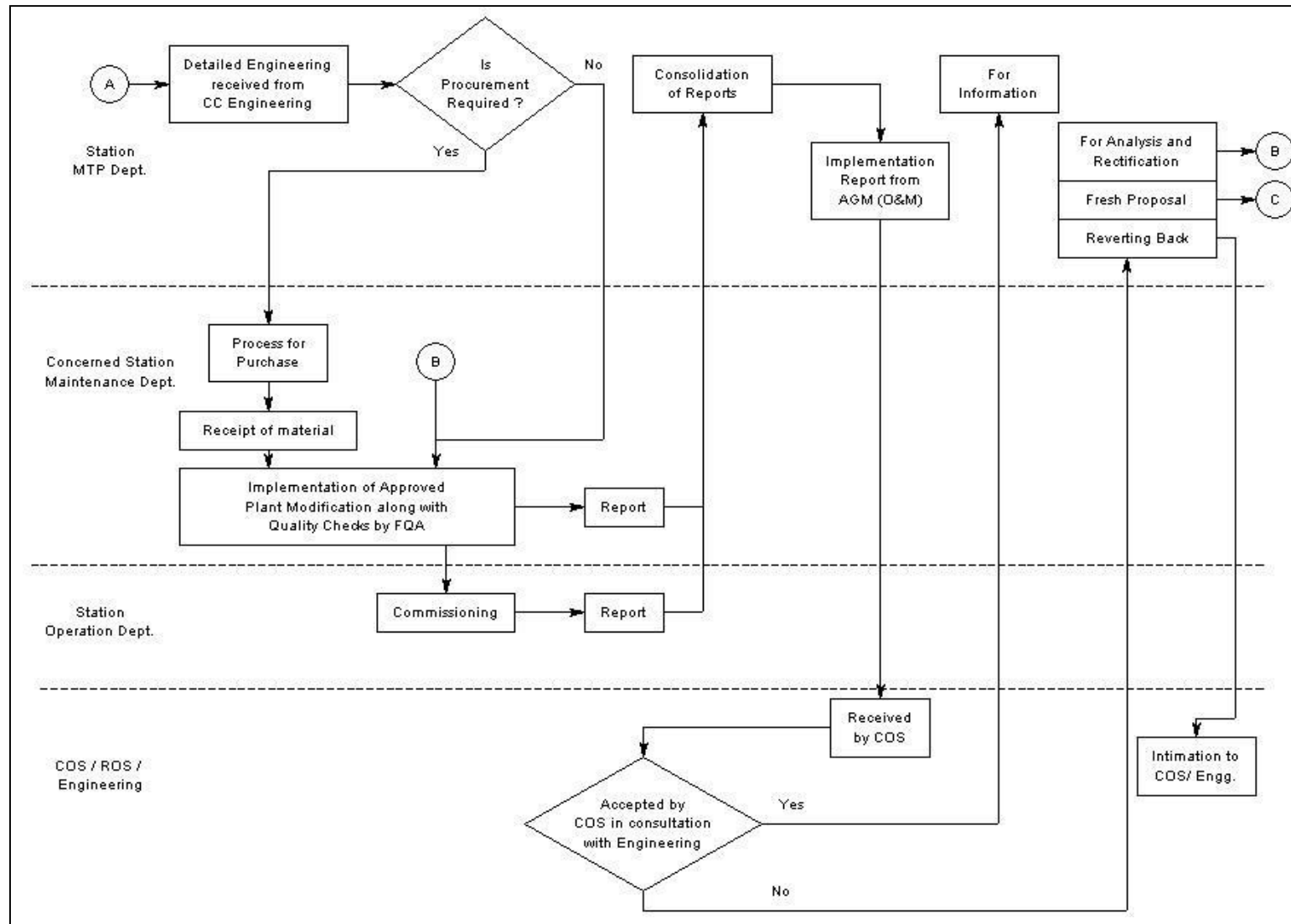
1. PROPOSED FOR MODIFICATION AT **POWER STATION**

Station Code:	Purpose of Modification (State justification)	
Component / System:		
Manufacturer :		
Details of Proposal (including time schedule): a) Whether the modification suggested is for phasing out an obsolete item, if so, details. b) Whether the modification suggested is for indigenisation, if so, details.	Effect on interchangeability	
Any identified safety hazard / Environment impact on account of the Plant Modifications		
Requirement for amending documents	Yes	No
Pre-modification Condition (e.g. Operational Performance, Emissions, Plant Life. etc.):		
Approval of Head of O&M	Date:	
Approval Head of Power Station	Date:	
2. REVIEW BY ED (REGION) & CATEGORIZATION		

Category of modification as per note ref. no. received from ROS		Date
3. APPROVALS		
Final approval from Engineering	Note Ref.	Date:
i) Not Approved	Note Ref. etc.	Date:
4. IMPLEMENTATION		
(After work has been completed, record the means by which modification has been achieved. State Contract Number, etc.)		Signature: Concerned Dept. I/C Date:
5. COMMISSIONING		
Confirm that commissioning has been done as per procedure and the relevant protocol; has been filled up.		Signature: Operation Dept. I/C Date:
6. AMENDMENTS TO DOCUMENTATION		
Confirm the completion of amendments to operating/maintenance Instructions, station manuals, plant status records, stores records, drawings, training programmes, etc.		Signature: Concerned Dept. I/C Date:
7. POST MODIFICATION DATA		
Date of modification	Signature of Initiator of Proposal: Date:	
Confirm completion of commissioning and state the condition after modification compared to the original data recorded in Section 1		
	Relevant Report Ref. etc	
8. CIRCULATION		
TO	DATE	SIGNATURE
Note:		
1. This LMI applies only to modifications; it does not apply to 'defects' or repairs which require equipment to be reinstated to the original condition.		
2. Section 2 'Categorization'-The selection of Assessment Category is the responsibility of the ED (Region)		

ANNEXURE – II

FLOW DIAGRAM FOR PROCESS OF MODIFICATION / DESIGN CHANGE



Annexure - II

FLOW DIAGRAM FOR PROCESS OF MODIFICATION / DESIGN CHANGE (CONTD.)

